

Communication lower bounds for Multi-Party Graph Traversal

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Abstract

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Motivation

- Figure of multiple agents on graph
- Point about good algorithm designers finding $\mathcal{O}$ bounds
- CC allows for finding Ω bounds, reducing problem scope for algorithm designers

Basic Example: Equality

There are two players, traditionally called **Alice** and **Bob**.

- Alice holds input x
- Bob holds input y
- Together, they want to compute a function $f(x, y)$

They are allowed to send messages (bits) back and forth. The goal is to minimize the total number of bits communicated.

Equality wishes to compute if two bit-strings are the same:

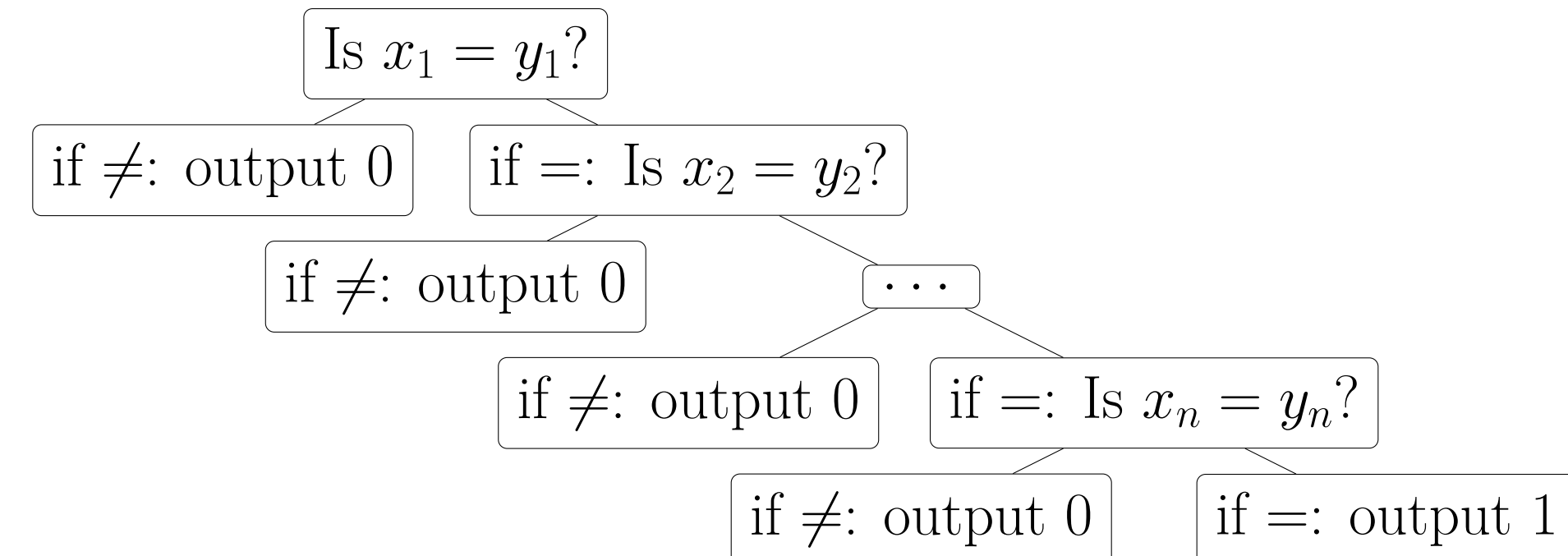


Figure 1: Visual for Equality Proof

Figure 3 demonstrates how, in the worst case, we must check every bit, since if every bit except x_n and y_n are the same, we will have to check all n bits to show equality. Thus, the deterministic communication complexity of Equality is $D(\text{EQ}) = n$

Two Player TRAV with EQ

The following materials were required to complete the research:

- Curabitur pellentesque dignissim
- Eu facilisis est tempus quis
- Duis porta consequat lorem
- Eu facilisis est tempus quis

The materials were prepared according to the steps outlined below:

- Curabitur pellentesque dignissim
- Eu facilisis est tempus quis
- Duis porta consequat lorem
- Curabitur pellentesque dignissim

Multi-Player TRAV, disjoint graphs

Nam quis odio enim, in molestie libero. Vivamus cursus mi at nulla elementum sollicitudin. Nam quis odio enim, in molestie libero. Vivamus cursus mi at nulla elementum sollicitudin.

$$E = mc^2 \quad (1)$$

Nam quis odio enim, in molestie libero. Vivamus cursus mi at nulla elementum sollicitudin. Nam quis odio enim, in molestie libero. Vivamus cursus mi at nulla elementum sollicitudin.

$$\cos^3 \theta = \frac{1}{4} \cos \theta + \frac{3}{4} \cos 3\theta \quad (2)$$

Nam quis odio enim, in molestie libero. Vivamus cursus mi at nulla elementum sollicitudin. Nam quis odio enim, in molestie libero. Vivamus cursus mi at nulla elementum sollicitudin.

$$\kappa = \frac{\xi}{E_{\max}} \quad (3)$$

Two Player TRAV with DISJ.

Lorem ipsum dolor **sit amet**, consectetur adipiscing elit. Sed laoreet accumsan mattis. Integer sapien tellus, auctor ac blandit eget, sollicitudin vitae lorem. Praesent dictum tempor pulvinar. Suspendisse potenti. Sed tincidunt varius ipsum, et porta nulla suscipit et. Etiam congue bibendum felis, ac dictum augue cursus a. **Donec** magna eros, iaculis sit amet placerat quis, laoreet id est. In ut orci purus, interdum ornare nibh. Pellentesque pulvinar, nibh ac malesuada accumsan, urna nunc convallis tortor, ac vehicula nulla tellus eget nulla. Nullam lectus tortor, *consequat tempor hendrerit* quis, vestibulum in diam. Maecenas sed diam augue.

Multi-Player TRAV, connected graphs



Figure 2: Figure caption

Nunc tempus venenatis facilisis. Curabitur suscipit consequat eros non porttitor. Sed a massa dolor, id ornare enim:

Treatments	Response 1	Response 2
Treatment 1	0.0003262	0.562
Treatment 2	0.0015681	0.910
Treatment 3	0.0009271	0.296

Table 1: Table caption

Two-player TRAV, iterative

Nunc tempus venenatis facilisis. **Curabitur suscipit** consequat eros non porttitor. Sed a massa dolor, id ornare enim. Fusce quis massa dictum tortor **tincidunt mattis**. Donec quam est, lobortis quis pretium at, laoreet scelerisque lacus. Nam quis odio enim, in molestie libero. Vivamus cursus mi at *nulla elementum sollicitudin*.

Multi-Player TRAV, iterative

Maecenas ultricies feugiat velit non mattis. Fusce tempus arcu id ligula varius dictum.

- Curabitur pellentesque dignissim
- Eu facilisis est tempus quis
- Duis porta consequat lorem

References

Acknowledgements

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